Experiment - 10

Aim : To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

Theory :

1. Introduction

Blockchain is a distributed digital ledger that stores information in a secure and transparent manner. The data is stored in blocks and shared across multiple computers in a network. Once a transaction is recorded and confirmed, it becomes difficult to modify without authorization and network agreement.

The **Blockchain-Based Lost and Found System for College** uses blockchain technology to maintain records of lost and found items within a college campus. It helps record item details, check existing records, and update the status of an item.

2. Hyperledger Fabric

Hyperledger Fabric is an open-source, permissioned blockchain platform designed for enterprise applications. A permissioned blockchain allows only authorized participants to access and participate in the network according to defined permissions.

In this experiment, Hyperledger Fabric is used to create the blockchain network, establish a channel, and deploy the chaincode that manages lost and found item records.

3. Components of Hyperledger Fabric

- **Peer:** A network node that hosts the ledger and executes chaincode.
- **Ordering Service:** Orders transactions and packages them into blocks.
- **Channel:** A private communication and ledger environment for participating organizations.
- **Ledger:** Stores the blockchain transaction history and the current state of the records.
- **Chaincode:** A program that defines the business logic and operations performed on the ledger.
- **Organizations:** Represent the participants that manage and operate the blockchain network.

4. Chaincode

Chaincode is the smart contract program used in Hyperledger Fabric. It defines how data is created, retrieved, and updated on the ledger.

In this experiment, a JavaScript chaincode named lostfound is deployed on the mychannel channel. It provides the following functions:

1. **ReportLostItem:** Records the details of a lost item.
2. **ReportFoundItem:** Records the details of a found item.
3. **GetItem:** Retrieves the details of a particular item using its ID.
4. **GetAllItems:** Retrieves all item records.
5. **UpdateItemStatus:** Updates the status of an existing item.

5. Working of the System

When a user reports a lost or found item, the chaincode receives the item's ID, name, location, and reporter's name. It creates a record and stores it on the blockchain ledger.

Each item has a status, such as LOST, FOUND, CLAIMED, or RETURNED. Users can query an item's details using its unique ID. The status can also be updated when the item is found or returned.

For example, a black wallet lost in the college library can be recorded with item ID item101 and status LOST. When the wallet is returned, its status can be updated to RETURNED.

Case Study : 'Blockchain-Based Lost and Found System for College'

Step 1: Update the System

```
admin27@exam-ca4-5:/mnt/c/Users/admin$ whoami
admin27
admin27@exam-ca4-5:/mnt/c/Users/admin$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [606 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [141 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [205 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [67.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [495 kB]
```

```
admin27@exam-ca4-5:/mnt/c/Users/admin$ sudo apt upgrade -y
Upgrading:
  bsdextrautils            gpg                      libgirepository-2.0-0  libpython3.14-mi
  bsdutils                 gpg-agent                libglib2.0-0t64        libpython3.14-st
  bubblewrap               gpg-wks-client           libglib2.0-bin         librsvg2-2
  ca-certificates          gpgconf                  libglib2.0-data        librsvg2-common
  cloud-init               gpgsm                    libglycin-2-0          libsmartcols1
  cloud-init-base          gpgv                     libheif-plugin-aomdec  libsqlite3-0
  curl                     libaom3                  libheif-plugin-aomenc  libssh2-1t64
  diffutils                libattr1                 libheif1               libssl3t64
  dirmngr                  libaudit-common          liblzma5               libsystemd-share
  distro-info-data         libaudit1                libmount1              libsystemd0
  eject                    libblkid1                libnetplan1            libudev1
  fdisk                    libbz2-1.0               libnss-systemd         libuuid1
  gir1.2-girepository-3.0  libc-bin                 libpam-modules         libx11-6
  gir1.2-glib-2.0          libc-gconv-modules-extra libpam-modules-bin     libx11-data
  git                      libc6                    libpam-runtime         libx11-xcb1
  git-man                  libcupl3t64-gnutls       libpam-systemd         libxml2-16
```

```
admin27@exam-ca4-5:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 16063, done.
remote: Counting objects: 100% (3470/3470), done.
remote: Compressing objects: 100% (641/641), done.
remote: Total 16063 (delta 3045), reused 2829 (delta 2829), pack-reused 12593 (from 3)
Receiving objects: 100% (16063/16063), 24.08 MiB | 18.05 MiB/s, done.
Resolving deltas: 100% (9276/9276), done.
admin27@exam-ca4-5:~$
```

```
admin27@exam-ca4-5:~$ cd ~/fabric-samples
admin27@exam-ca4-5:~/fabric-samples$ ls
CHANGELOG.md        MAINTAINERS.md        asset-transfer-ledger-queries     auction-simple                  off_chain_data          token-erc-1155
CODEOWNERS          README.md             asset-transfer-private-data       ci                              test-application        token-erc-20
CODE_OF_CONDUCT.md  asset-transfer-abac   asset-transfer-sbe                full-stack-asset-transfer-guide test-network            token-erc-721
CONTRIBUTING.md     asset-transfer-basic  asset-transfer-secured-agreement  hardware-security-module        test-network-k8s        token-sdk
LICENSE             asset-transfer-events auction-dutch                     high-throughput                 test-network-nano-bash  token-utxo
admin27@exam-ca4-5:~/fabric-samples$
```

```
admin27@exam-ca4-5:~/fabric-samples$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
admin27@exam-ca4-5:~/fabric-samples$ chmod +x install-fabric.sh
admin27@exam-ca4-5:~/fabric-samples$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.16 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
===> Will unpack to: /home/admin27/fabric-samples
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0                              0
100 121.8M 100 121.8M   0      0  2.10M      0   00:57   00:57          5.54M
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
===> Will unpack to: /home/admin27/fabric-samples
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0                              0
100 31.65M 100 31.65M   0      0  3.22M      0   00:09   00:09          2.60M
==> Done
```

```
admin27@exam-ca4-5:~/fabric-samples$ peer version
peer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

```
/home/admin27/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/my
2026-10-26 04:07:10.591 UTC 0001 INFO [common.tools.configtxgen] main -> Lo
2026-10-26 04:07:10.595 UTC 0002 INFO [common.tools.configtxgen.localconfig
2026-10-26 04:07:10.595 UTC 0003 INFO [common.tools.configtxgen.localconfig
election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_s
2026-10-26 04:07:10.595 UTC 0004 INFO [common.tools.configtxgen.localconfig
aml
2026-10-26 04:07:10.597 UTC 0005 INFO [common.tools.configtxgen] doOutputBl
2026-10-26 04:07:10.597 UTC 0006 INFO [common.tools.configtxgen] doOutputBl
2026-10-26 04:07:10.598 UTC 0007 INFO [common.tools.configtxgen] doOutputBl
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
        "name": "mychannel",
        "url": "/participation/v1/channels/mychannel",
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}
```

```
2026-10-26 04:09:09.923 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [316et
 at localhost:7051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc,
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc,
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$
admin27@exam-ca4-5:~/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-samples/bin
admin27@exam-ca4-5:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
admin27@exam-ca4-5:~/fabric-samples/test-network$ export CORE_PEER_TLS_ENABLED=true
admin27@exam-ca4-5:~/fabric-samples/test-network$ export CORE_PEER_LOCALMSPID="Org1MSP"
admin27@exam-ca4-5:~/fabric-samples/test-network$ export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-netu
peers/peer0.org1.example.com/tls/ca.crt
admin27@exam-ca4-5:~/fabric-samples/test-network$ export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/
s/Admin@org1.example.com/msp
admin27@exam-ca4-5:~/fabric-samples/test-network$ export CORE_PEER_ADDRESS=localhost:7051
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAs
[]
admin27@exam-ca4-5:~/fabric-samples/test-network$
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode invoke \
>   -o localhost:7050 \
>   --ordererTLSHostnameOverride orderer.example.com \
>   --tls \
>   --cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tl
>   -C mychannel \
>   -n basic \
>   --peerAddresses localhost:7051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.exampl
>   --peerAddresses localhost:9051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.exampl
>   -c '{"function":"InitLedger","Args":[]}'
2026-10-26 04:11:52.049 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size"
:5,"docType":"asset"},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yellow","ID":"as
set4","Owner":"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":80
0,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
```

```
admin27@exam-ca4-5:~/fabric-samples/college-lost-found/chaincode$ cat package.json

{
  "name": "college-lost-found",
  "version": "1.0.0",
  "description": "Blockchain-Based Lost and Found System for College",
  "main": "lostfound.js",
  "scripts": {
    "start": "fabric-chaincode-node start"
  },
  "engines": {
    "node": ">=18"
  },
  "dependencies": {
    "fabric-contract-api": "^2.5.8",
    "fabric-shim": "^2.5.8"
  }
}
```

```
admin27@exam-ca4-5:~/fabric-samples/college-lost-found/chaincode$ npm install

added 79 packages, and audited 80 packages in 8s

6 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ ./network.sh deployCC -ccn lostfound -ccp ../college-lost-found/chaincode -ccl javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: lostfound
- CC_SRC_PATH: ../college-lost-found/chaincode
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: lostfound
- CC_SRC_PATH: ../college-lost-found/chaincode
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package lostfound.tar.gz --path ../college-lost-found/chaincode --lang node --label lostfound_1.0
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode invoke \
>   -o localhost:7050 \
>   --ordererTLSHostnameOverride orderer.example.com \
>   --tls \
>   --cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem" \
>   -C mychannel \
>   -n lostfound \
>   --peerAddresses localhost:7051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
>   --peerAddresses localhost:9051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
>   -c '{"function":"ReportLostItem","Args":["item101","Black Wallet","College Library","Himanshi"]}'
2026-10-26 04:20:53.990 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"{\"itemId\":\"item101\",\"itemNa
me\":\"Black Wallet\",\"location\":\"College Library\",\"reporter\":\"Himanshi\",\"status\":\"LOST\"}"
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n lostfound -c '{"Args":["GetItem","item101"]}'
{"itemId":"item101","itemName":"Black Wallet","location":"College Library","reporter":"Himanshi","status":"LOST"}
admin27@exam-ca4-5:~/fabric-samples/test-network$
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode invoke \
>   -o localhost:7050 \
>   --ordererTLSHostnameOverride orderer.example.com \
>   --tls \
>   --cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlsc
>   -C mychannel \
>   -n lostfound \
>   --peerAddresses localhost:7051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.
>   --peerAddresses localhost:9051 \
>   --tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.
>   -c '{"function":"UpdateItemStatus","Args":["item101","RETURNED"]}'
2026-10-26 04:22:33.089 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 paylo
me\":\"Black Wallet\",\"location\":\"College Library\",\"reporter\":\"Himanshi\",\"status\":\"RETURNED\"}"
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n lostfound -c '{"Args":["GetItem","item101"]}'
{"itemId":"item101","itemName":"Black Wallet","location":"College Library","reporter":"Himanshi","status":"RETURNED"}
```

```
admin27@exam-ca4-5:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n lostfound -c '{"Args":["GetAllItems"]}'
[{"itemId":"item101","itemName":"Black Wallet","location":"College Library","reporter":"Himanshi","status":"RETURNED"},{"itemId":"item102","itemName":"Blue Water Bottle
","location":"College Canteen","reporter":"Rahul","status":"FOUND"}]
```

Implementation Summary

The Blockchain-Based Lost and Found System for College was implemented using Hyperledger Fabric. A blockchain network was configured with two organizations and a channel named mychannel. A custom JavaScript chaincode named lostfound was deployed on the channel.

The chaincode supports the following operations:

1. **ReportLostItem:** Records details of a lost item.
2. **ReportFoundItem:** Records details of a found item.
3. **GetItem:** Retrieves the details of a specific item.

4. **GetAllItems:** Retrieves all recorded items.
5. **UpdateItemStatus:** Updates an item's status to LOST, FOUND, CLAIMED, or RETURNED.

The implementation was tested by recording a lost wallet, recording a found water bottle, querying item details, and updating the wallet's status. Successful query results demonstrate that the chaincode can store and retrieve item records through the Hyperledger Fabric network.

Result: The basic blockchain-based Lost and Found system was implemented and tested successfully, subject to the successful execution of the above transactions.

6. Advantages

- Provides a shared and traceable record of item information.
- Makes it easier to retrieve records using an item ID.
- Supports controlled updates through chaincode functions.
- Maintains transaction history on the blockchain.
- Reduces dependence on a single centralized record system.

7. Applications

This system can be used in colleges, schools, libraries, hostels, offices, and other public places to manage lost and found items.

8. Conclusion

The experiment demonstrates the implementation of a basic Blockchain-Based Lost and Found System using Hyperledger Fabric. A blockchain network and channel were created, and custom JavaScript chaincode was deployed. The chaincode supports recording lost and found items, retrieving item details, viewing all records, and updating item status. This system demonstrates how blockchain can be applied to maintain traceable and shared records for lost and found management.